

The Hardware Barrier to Over-the-Air Adversarial RF-Fingerprint Spoofing: Why External Radiation Fails but Transmitter Pre-Distortion Succeeds

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Abstract

Radio-frequency device-fingerprinting classifiers identify a transmitter from the hardware impairments—power-amplifier nonlinearity, I/Q imbalance, carrier frequency offset, phase noise—stamped on its emissions. We ask whether a digital adversarial perturbation can spoof such a classifier over the air, and give a structural answer that turns on *whose hardware the perturbation passes through*. A targeted perturbation that flips the classifier from a legitimate device to a chosen target with probability one at the receiver input yields $\approx 0\%$ fooling when it is radiated by a *second, external* transmitter: the classifier reads the emitter’s hardware parameters, but every signal the adversary radiates is passed through the adversary’s own—foreign and non-invertible—transmit operator, so it carries the *adversary’s* fingerprint, not the target’s. This barrier is strictly deeper than a channel-limited attack— expectation-over-transformation (channel-aware) crafting neutralizes the propagation channel but not the adversary’s hardware operator, and over-the-air fooling stays $\approx 0\%$. By contrast, when the *same* perturbation is pre-distorted into the legitimate transmitter’s own baseband—riding through its *invertible* hardware operator—the target *is* reachable: the required drive exists in closed form, and experimentally the classifier reads the target for 71% of frames at unit perturbation-to-signal ratio (versus 0% for a random perturbation of equal power, and 0% for the external adversary at every power). The hardware barrier therefore stops an external radiator but not pre-distortion at the legitimate transmitter—the decisive distinction being *addition through a foreign non-invertible operator* versus *pre-composition with an invertible own operator*.

We formalize both halves of this result: why a targeted perturbation δ that flips a device-fingerprint classifier from a legitimate device to a target with probability 1 at the receiver input yields ≈ 0 fooling when radiated by a *second* transmitter (the hardware barrier, deeper than Kim’s channel-limited result), and why the same δ *does* reach the target when pre-distorted through the legitimate transmitter’s own invertible hardware.

Notation

<i>Signals and hardware</i>	
$n, u[n]$	sample index; intended baseband samples (digital content)
$\mathcal{T}_d$	transmit (hardware) operator of device d , eq. (1)
$g_d, M_d, c_d, \omega_d, \xi_d$	PA nonlinearity, I/Q imbalance, DC offset, CFO, phase noise of d
$\boldsymbol{\theta}_d$	fingerprint: impairment vector $(g_d, M_d, c_d, \omega_d, \xi_d)$
h_d, w, y	channel $d \rightarrow \text{RX}$; additive noise; received signal, eq. (3)
<i>Classifier</i>	
f	fingerprint classifier
$s_k(y)$	score (logit) for device k ; decision $\hat{d}(y) = \arg \max_k s_k$
$S_k, \hat{\boldsymbol{\theta}}(y)$	score vs. impairment params; signature extracted from y
<i>Attack</i>	
x	captured device 6 frame ($f(x) = \text{device 6}$); attack start point
δ, δ^*	perturbation; PGD (digital) optimum
δ'	digital drive loaded into the adversary's DAC
$\varepsilon, \mathbf{e}_4, \text{CE}$	budget $\ \delta\ \leq \varepsilon \ x\ $; target one-hot (device 4); cross-entropy
device 6 / device 4	legit transmitter / spoofing target
<i>Single-channel (pre-distortion)</i>	
ℓ, τ	generic legitimate transmitter / target device (expt: dev 6 / dev 4)
$y_{\text{SC}}, J_\ell(u)$	single-channel emission eq. (13); Jacobian of $\mathcal{T}_\ell$ at u , eq. (14)
$\mathcal{P}_\ell$	emittable perturbation set through own hardware (full rank)
<i>Over-the-air and EOT</i>	
subscript a	adversary radio: $\mathcal{T}_a, h_a, \boldsymbol{\theta}_a$
$y_{\text{OTA}}, \mathcal{P}_a$	OTA composite eq. (8); physically radiable set
PSR	perturbation-to-signal ratio (dB)
$t, \mathcal{T}$	a transformation and its distribution (EOT)
$h, \mathcal{H}$	a channel realization and its distribution (EOT)
δ_{EOT}^*	EOT (distribution-robust) perturbation

1 Hardware transmit operator (the fingerprint)

Each device d maps intended baseband samples $u[n] \in \mathbb{C}$ to an emitted RF baseband signal through a nonlinear transmit operator $\mathcal{T}_d$:

$$\mathcal{T}_d(u)[n] = g_d(M_d u[n] + c_d) e^{j\omega_d n} + \xi_d[n], \quad (1)$$

with power-amplifier nonlinearity $g_d(\cdot)$ (AM/AM + AM/PM, *non-invertible past saturation*), I/Q imbalance M_d , DC offset c_d , carrier frequency offset ω_d , and phase noise ξ_d . Collect the impairment parameters as the *fingerprint*

$$\boldsymbol{\theta}_d = (g_d, M_d, c_d, \omega_d, \xi_d). \quad (2)$$

2 Received signal and classifier

Through channel h_d with noise w ,

$$y = h_d * \mathcal{T}_d(u) + w. \quad (3)$$

The classifier f is trained (per-window unit-RMS normalization; CFO/phase/time augmentation) to be invariant to content u and channel h and to read the impairments. Hence its class scores factor through an extracted signature $\hat{\boldsymbol{\theta}}(y)$:

$$s_k(y) \approx S_k(\hat{\boldsymbol{\theta}}(y)), \quad \hat{d}(y) = \arg \max_k s_k(y). \quad (4)$$

Key: f is a function of the hardware parameters θ , not of the raw content.

3 Digital targeted PGD (construction of δ)

Take a captured device 6 frame x with $f(x) = \text{dev } 6$ (i.e. $\hat{\theta}(x) \approx \theta_6$). Solve, over an ℓ_2 ball of radius $\varepsilon\|x\|$,

$$\delta^* = \arg \min_{\|\delta\| \leq \varepsilon\|x\|} \text{CE}(f(x + \delta), e_4). \quad (5)$$

PGD ascends $\nabla_{\delta} s_4(x + \delta)$ *through the classifier*, so by construction

$$\delta^* \in \text{input (signal) space of } f, \text{ defined by } \nabla f, \text{ such that } S_4(\hat{\theta}(x + \delta^*)) > S_6(\hat{\theta}(x + \delta^*)). \quad (6)$$

Empirically $f(x + \delta^*) = \text{dev } 4$ with probability 1.

4 Digital evaluation succeeds

Digitally, $x + \delta^*$ is fed *directly* to f ; nothing sits between δ^* and the input:

$$f(x + \delta^*) = \text{dev } 4. \quad \checkmark \quad (7)$$

5 Over-the-air (two-channel) signal

Device 6 emits the legit frame $h_6 * \mathcal{T}_6(u)$ (carries θ_6); the adversary loads a digital drive δ' into *its* DAC and emits $h_a * \mathcal{T}_a(\delta')$ (carries θ_a). They superimpose linearly in the air:

$$y_{\text{OTA}} = h_6 * \mathcal{T}_6(u) + h_a * \mathcal{T}_a(\delta') + w. \quad (8)$$

We require $f(y_{\text{OTA}}) = \text{dev } 4$.

6 The exact condition, and why it cannot hold

To reproduce the digital success the radiated perturbation must equal δ^* :

$$h_a * \mathcal{T}_a(\delta') \stackrel{!}{=} \delta^* \implies \delta' = \mathcal{T}_a^{-1}(h_a^{-1}(\delta^*)). \quad (9)$$

This is infeasible: (i) $\mathcal{T}_a^{-1}$ does not exist (g_a non-invertible past saturation; δ^* may lie outside $\text{range}(\mathcal{T}_a)$); (ii) h_a is unknown without channel awareness; (iii) every physically radiated signal is passed through $\mathcal{T}_a$, which *imprints θ_a on everything it emits*.

7 Reachable-set argument (core result)

Define the set of perturbations the adversary can physically place on the air:

$$\mathcal{P}_a = \{ h_a * \mathcal{T}_a(\delta') : \|\delta'\| \leq \varepsilon \}, \quad \text{every element carries } \theta_a. \quad (10)$$

The OTA attack succeeds iff $\exists p \in \mathcal{P}_a$ with $f(h_6 * \mathcal{T}_6(u) + p) = \text{dev } 4$. Because f reads impairments and every $p \in \mathcal{P}_a$ carries θ_a , the composite signature is a mixture along the θ_6 – θ_a axis parameterized by the perturbation-to-signal ratio (PSR):

$$\hat{\theta}(h_6 * \mathcal{T}_6(u) + p) = \Theta(\theta_6, \theta_a; \text{PSR}). \quad (11)$$

As PSR grows, the reading slides $\text{dev } 6 \rightarrow (\text{device with fingerprint } \approx \boldsymbol{\theta}_a)$, and

$$\boxed{f(y_{\text{OTA}}) = \text{dev } 4 \iff \boldsymbol{\theta}_a \approx \boldsymbol{\theta}_4} \quad (12)$$

i.e. only if the adversary radio *is* device 4. The digital solution δ^* is a direction in ∇f -space; $\mathcal{P}_a$ is a manifold in hardware- $\boldsymbol{\theta}_a$ -space; generically $\delta^* \notin \mathcal{P}_a$. This is the observed 0%.

8 Single-channel pre-distortion: the target *is* reachable

The barrier above is specific to a *foreign* radiator. Consider instead a *single-channel* (pre-distortion) attack in which the drive δ is injected into the legitimate transmitter's baseband, *upstream* of its own hardware. Write ℓ for the legitimate device and τ for the target device (in our experiments $\ell = \text{dev } 6$, $\tau = \text{dev } 4$). The emitted signal is

$$y_{\text{SC}} = h_\ell * \mathcal{T}_\ell(u + \delta) + w. \quad (13)$$

Now δ is *pre-composed* with $\mathcal{T}_\ell$: it passes through the *same* operator as the legitimate signal, not through a foreign $\mathcal{T}_a$. The perturbation it places on the emission is the operator difference

$$e(\delta) = \mathcal{T}_\ell(u + \delta) - \mathcal{T}_\ell(u) \approx J_\ell(u) \delta, \quad J_\ell(u) = \text{diag}(g'_\ell(M_\ell u + c_\ell)) M_\ell e^{j\omega_\ell n}, \quad (14)$$

where $J_\ell(u)$ is the Jacobian of $\mathcal{T}_\ell$ at u . Wherever the PA is not in hard saturation ($g'_\ell \neq 0$) and the I/Q mixing M_ℓ is non-singular, $J_\ell(u)$ has *full rank* and is invertible — in contrast to $\mathcal{T}_a$, which is foreign and non-invertible past saturation.

Reachable set. The physically emittable perturbations are

$$\mathcal{P}_\ell = \{ \mathcal{T}_\ell(u + \delta) - \mathcal{T}_\ell(u) : \|\delta\| \leq \varepsilon \|u\| \} \approx J_\ell(u) \cdot \text{Ball}(\varepsilon \|u\|). \quad (15)$$

Because $J_\ell(u)$ is full rank, $\mathcal{P}_\ell$ is a *full-dimensional* neighbourhood of the origin in emission space: it spans every direction as ε grows. In particular the emission-domain direction $d^* = h_\ell * \delta^*$ that the classifier reads as device τ lies in $\mathcal{P}_\ell$ for large enough ε , and the required drive *exists* in closed form (to first order),

$$\delta^* \approx J_\ell(u)^{-1} d^* = J_\ell(u)^{-1} (h_\ell * \delta_{\text{dig}}^*), \quad (16)$$

in sharp contrast to the two-channel condition (9), where $\delta' = \mathcal{T}_a^{-1}(h_a^{-1}(\delta^*))$ does *not* exist. The whole distinction is *pre-composition with an invertible own-hardware operator* versus *addition through a foreign non-invertible one*:

$$\underbrace{\mathcal{T}_\ell(u + \delta)}_{\delta \text{ inside } \mathcal{T}_\ell \text{ (invertible)}} \quad \text{vs.} \quad \underbrace{\mathcal{T}_\ell(u) + h_a * \mathcal{T}_a(\delta')}_{\delta' \text{ through foreign } \mathcal{T}_a}. \quad (17)$$

Hence there is a finite drive at which the reading crosses into device τ :

$$\boxed{\exists \varepsilon^* : \quad f(h_\ell * \mathcal{T}_\ell(u + \delta^*)) = \text{device } \tau, \quad \|\delta^*\| = \varepsilon^* \|u\|.} \quad (18)$$

Experimental confirmation. With correctly scaled δ (realized PSR = label) passed through the legitimate transmitter's own hardware, the reading flips to the target at ε^* corresponding to PSR ≈ 0 dB ($\|\delta\| \approx \|u_{\text{payload}}\|$): $f(y_{\text{SC}}) = \text{device } \tau$ for 71% of frames, versus 0% for a *random* δ of equal power (ruling out loud-signal collapse) and 0% for the two-channel adversary at every PSR. The barrier of (9)–(18) therefore holds for an *external radiator* ($\mathcal{T}_a$) but *not* for pre-distortion at the legitimate transmitter, whose own invertible $\mathcal{T}_\ell$ the drive rides through.

9 Contrast with Kim (channel-limited attack)

If instead the classifier reads *content* (modulation), $s_k(y) \approx S_k(c(y))$, the perturbation acts on content, which traverses the linear part and is channel-correctable. Making δ channel-aware [2] handles h_a^{-1} and the perturbation reaches the content decision region: the attack *works*. There the sole barrier is the channel h_a . Here the barrier is the hardware operator $\mathcal{T}_a$, which no digital drive from a foreign radio can invert, because the target θ_4 is a property of the *emitter*, not of the injected signal.

10 Prediction for Expectation-over-Transformation (EOT)

EOT [1] robustifies over the channel,

$$\delta_{\text{EOT}}^* = \arg \max_{\|\delta\| \leq \varepsilon \|x\|} \mathbb{E}_{h \sim \mathcal{H}} [s_4(h(x + \delta))], \quad (19)$$

which neutralizes h_a —the whole of Kim’s barrier.¹ But EOT does nothing to $\mathcal{T}_a$: the emitted signal is still $\mathcal{T}_a(\delta_{\text{EOT}}')$ and still carries θ_a . Therefore

$$\text{Kim's barrier } (h_a) : \text{ closed by EOT; } \quad \text{hardware barrier } (\mathcal{T}_a) : \text{ untouched by EOT.} \quad (20)$$

Prediction (confirmed in §11): EOT remains $\approx 0\%$ toward device 4 unless $\theta_a \approx \theta_4$. If the EOT run still fails, the barrier is hardware ($\mathcal{T}_a$), strictly deeper than Kim’s channel result; if it succeeds, the limitation was only the channel, i.e. the same as Kim.

11 Experimental confirmation

Session-14 over-the-air measurements (hardware-fingerprint CVNN; device 6 legit, device 4 target; targeted PGD; δ confined to the payload; pristine $\mathcal{T}_6(u)$ on ch0 so device 6 stamps its fingerprint once). The classifier scoring the captures is domain-adapted to the replay so the δ -off baseline reads device 6.

Setting	targeted $\rightarrow$ device 4
Digital (δ at receiver input)	100%
OTA, no-EOT (2-channel)	$\approx 0\%$ (all PSR)
OTA, channel-aware EOT (2-channel)	$\approx 0\%$ (all PSR)

Per-PSR fraction (channel-aware EOT sweep, replay-adapted model):

PSR (dB)	$\rightarrow$ device 4	$\rightarrow$ device 6
0 (loudest δ)	0.00	1.00
−5	0.04	0.75
−10	0.00	0.96
−15	0.04	0.95
−20	0.01	0.96
−30 ($\approx \delta$ -off)	0.21 [†]	0.68

¹EOT (Expectation over Transformation) is *not* channel estimation. Channel estimation *measures* the specific channel h_a and inverts it ($\delta' = h_a^{-1}(\delta^*)$); it needs pilots/feedback on that link. EOT instead maximizes the *expected* objective over a modeled *distribution* of channel realizations $\mathcal{H}$ (here random sub-sample time-shift and carrier-phase rotation), producing a δ robust to the whole family *without* estimating any single channel. We use EOT because the adversary $\rightarrow$ receiver link carries no pilot/feedback, so h_a is unobservable. Both, however, only address h_a ; neither touches $\mathcal{T}_a$.

[†] residual baseline mis-read noise, *anti-correlated* with δ power (largest where δ is weakest); a thr=2 extraction merged the near- δ -off capture into 16 s bursts and spuriously reported 0.60, corrected to 0.21 with thr ≥ 4 .

At the loudest perturbation (PSR 0 dB) the composite still reads device 6 with probability 1.00, and the fooling rate toward device 4 does *not* increase with perturbation power. Both the naive and the channel-aware over-the-air attacks give $\approx 0\%$ toward device 4. This confirms the prediction: EOT neutralizes the channel h_a but not the hardware operator $\mathcal{T}_a$, so the barrier is hardware — strictly deeper than Kim’s channel-limited result.

References

- [1] A. Athalye, L. Engstrom, A. Ilyas, and K. Kwok, “Synthesizing Robust Adversarial Examples,” in *Proc. 35th Int. Conf. on Machine Learning (ICML)*, 2018, pp. 284–293.
- [2] B. Kim, Y. E. Sagduyu, K. Davaslioglu, T. Erpek, and S. Ulukus, “Channel-Aware Adversarial Attacks Against Deep Learning-Based Wireless Signal Classifiers,” *IEEE Trans. Wireless Commun.*, vol. 21, no. 6, pp. 3868–3880, 2022.